Computer Networks

Lecture 23 Network Layer

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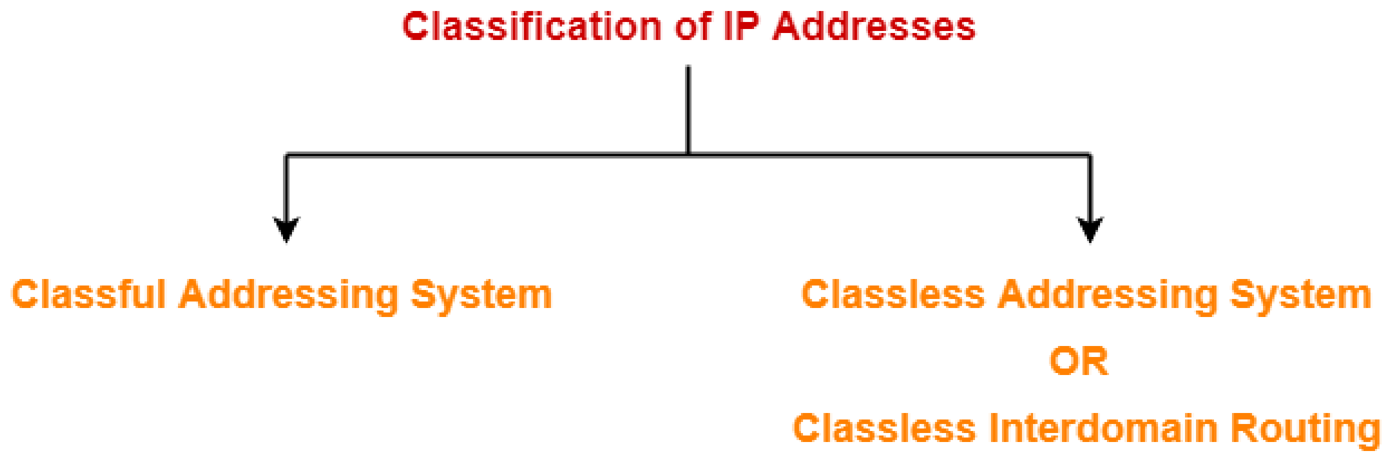
Content

- **Classless Addressing**

Introduction

To overcome address depletion and give more organizations access to the Internet, classless addressing was designed and implemented.

In this scheme, there are no classes, but the addresses are still granted in blocks.



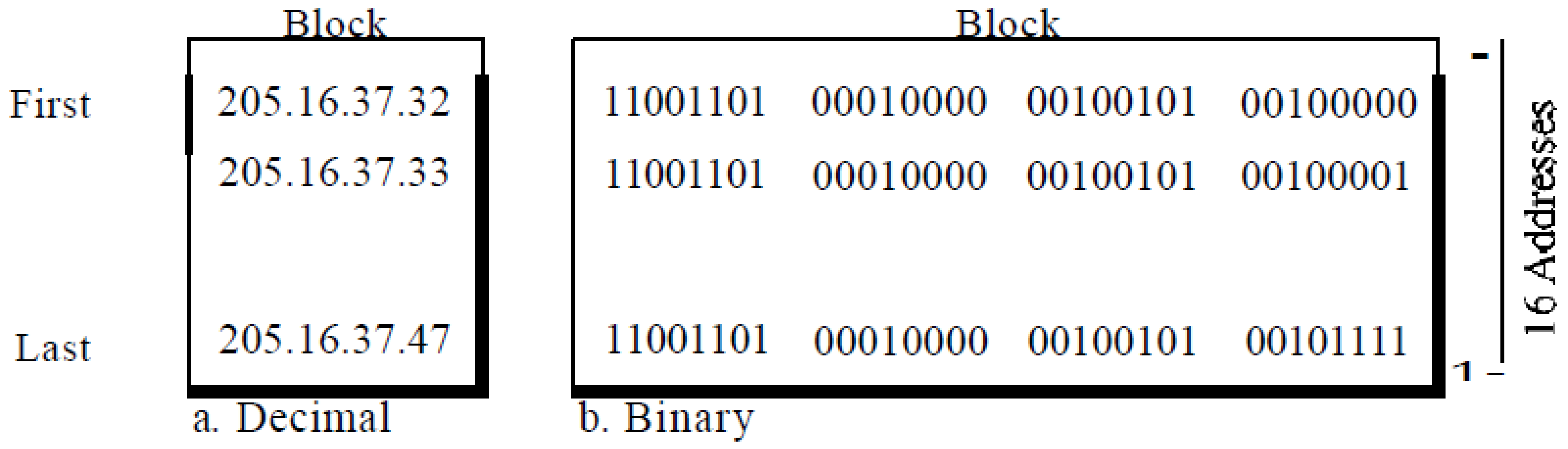
Address Blocks

In classless addressing, when an entity, small or large, needs to be connected to the Internet, it is granted a block (range) of addresses.

To simplify the handling of addresses, the Internet authorities impose three restrictions on classless address blocks:

1. The addresses in a block must be contiguous, one after another.
2. The number of addresses in a block must be a power of 2 (I, 2, 4, 8, ...).
3. The first address must be evenly divisible by the number of addresses.

Address Blocks: Example



1. The addresses are contiguous.
2. The number of addresses is a power of 2 (16 = 24).
3. the first address is divisible by 16.

Mask

A mask is a 32-bit number in which the n leftmost bits are $1s$ and the $32 - n$ rightmost bits are $0s$. However, in classless addressing the mask for a block can take any value from 0 to 32.

In 1Pv4 addressing, a block of addresses can be defined as x.y.z.t/n

in which x.y.z.t defines one of the addresses and the /n defines the mask.

The first address in the block can be found by setting the rightmost 32 - n bits to $0s$.

The last address in the block can be found by setting the rightmost 32 - n bits to $1s$.

Question

1. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address, last address and number of address in the block?
2. One of the address in the block is 167.199.170.82/27. Find the no. of addresses, 1st address and last address.
3. One of the address in the block is 17.63.110.114/24. Find the no. of addresses, 1st address and last address.
4. One of the address in the block is 110.23.120.14/20. Find the no. of addresses, 1st address and last address.
5. An organization is granted a block of address with the beginning address 14.24.74.0/24. The organization needs to have 3 sub-block of address to use in its 3 subnets as follows:
   a) One subblock of 120 address
   b) Another sub-block of 60 address
   c) Another subblock of 10 address